

Component Scores Do Not Compose: Clause Detection, Span Extraction and Category-Level Risk Prioritization on the Contract Understanding Atticus Dataset

Jerin Aktar, Afnan Mazumdar, Shoyeb Hasan Sayem and Utsha Kumar Roy

Abstract—Automated contract review is normally evaluated one component at a time, and the resulting scores are read as though they describe the assembled system. This work reports a three-stage pipeline on the Contract Understanding Atticus Dataset (CUAD; 510 contracts, 41 clause categories) under a strict contract-level 80/20 split, and shows that its component scores overstate delivered performance by a factor of three. A sliding-window DistilBERT trained under multiple-instance learning loses to a full-document TF-IDF baseline at clause presence classification (micro-F1 0.694 against 0.775). Four alternative explanations are tested and eliminated: training on all 53,662 window examples rather than a 24,000 subsample moves micro-F1 by -0.0004 ; three random seeds place the gap at 9.5 standard deviations; six aggregation rules at their own optimal thresholds all remain behind the baseline; and an untuned decision threshold accounts for 0.056 of the difference. A conjunctive ensemble reaches micro-F1 0.776, but its $+0.0014$ margin is smaller than the bootstrap interval, the model-seed variance and the dataset-split variance alike. Run end to end, only 21.7% (95% CI [19.6, 23.9]) of gold clauses survive all three stages against the 64% the component scores predict, and the loss is localized to window selection (31.7%) and a train/inference distribution mismatch. Finally, aggregate micro-F1 and the system’s stated purpose diverge: the configuration ranked first misses 37.5% of high-risk clauses while a configuration ranked last misses 13.6%. We argue that pipeline-level survival and a purpose-weighted metric should be reported alongside component scores as a matter of routine.

Index Terms—Legal natural language processing, contract analysis, CUAD, multiple-instance learning, model evaluation, calibration, risk prioritization, held-out evaluation.

I. Introduction

COMMERCIAL contracts routinely run to tens of thousands of characters, while the provisions that determine a signer’s exposure occupy a few hundred. In CUAD [1] the median contract is 33,143 characters and the median annotated clause is 196, so the document exceeds a standard transformer window by roughly $16\times$ while the target sits two orders of magnitude below it. CUAD frames the resulting task as extraction: given a category, locate the clause. It stops there, offering no indication of what a located clause means for the party signing it.

This paper reports a system that attempts the further step and, more usefully, an evaluation of where that system fails. Our contributions are:

- 1) A negative result on long-document presence classification, with four alternative explanations tested and eliminated (Section V-B).
- 2) A demonstration that stage-wise scores do not compose: 21.7% end-to-end survival against 64% predicted, decomposed into two identifiable causes (Section V-F).
- 3) Evidence that aggregate micro-F1 and user harm can invert the model ranking, measured through a risk-weighted evaluation (Section V-C).
- 4) A demonstration that a small reference set converts a generation task into classification without the metric registering the change (Section V-E).

II. Related Work

CUAD [1] supplies 510 commercial contracts with expert annotations over 41 clause categories, posed in SQuAD format [2]. Domain-adapted encoders [3] and benchmark suites [4] have advanced legal language understanding, but they operationalize importance as category membership rather than as severity for a particular party. We are not aware of an established risk-severity model covering CUAD’s categories, which is why the layer added here is presented as a prototype rather than as a validated instrument.

Long inputs are handled by retrieval, by windowing with aggregation, or by sparse attention [5]. Windowing aggregated with a maximum is the classical multiple-instance learning (MIL) rule [6]. For generation, T5 [7] and its instruction-tuned variant [8] express clause simplification naturally; Manor and Li [9] assemble plain-English contract summaries and observe a tendency toward generic output. A substantial literature establishes that fluent abstractive output is frequently unfaithful [10], [11] and that overlap metrics such as ROUGE [12] correlate poorly with faithfulness judgements; embedding metrics [13] improve similarity estimation without testing factuality. The summarization result reported here is an extreme instance of the same phenomenon, and the calibration analysis follows [14].

Two properties of CUAD dominate any modelling effort on it. The first is class imbalance: per-category positive rates span 100% to 2.5%, and the median category appears in roughly 23% of contracts. The second is length, which

forces every system to commit to a length strategy before it commits to a model. That choice is usually asserted; here it is measured, both as a retrieval ceiling obtained before training and as a cost benchmark for sparse attention on the target hardware.

III. Proposed Methodology

A. Pipeline Overview

The system is a staged pipeline. Contract text is windowed; a presence stage decides which of the 41 categories the contract contains; a span stage extracts the clause text for each detected category; a risk layer attaches the detected category’s level; and a text-generation stage attaches a plain-English sentence. Each stage is trained independently and, in conventional practice, would be evaluated independently.

B. Stage 1: Presence Classification

A TF-IDF representation over the entire document feeds per-category logistic regression, giving a baseline with no length limit. In parallel, 2,000-character windows at a 1,500-character stride tile each contract and a DistilBERT [15] scores (category question, window) pairs; the document-level score is the maximum over windows, which is the MIL decision rule. The 500-character overlap exceeds the median clause length, so no clause can fall between two windows. Parameter-free AND, OR and AVG rules combine the two models; none contains anything fittable, so the test set cannot tune them.

Before any training we measured the ceiling a retrieval design would impose. The best query strategy places the gold clause inside its selected windows only 64.0% of the time, below what the baseline already achieves, so retrieval was rejected on a measurement rather than an assumption. The scope of that claim is narrow and worth stating: it bounds lexical retrieval only, and a dense retriever may well raise it.

C. Stage 2: Span Extraction

A DistilBertForQuestionAnswering head predicts start and end token positions. At training time each example is re-centred so the answer begins approximately 150 characters into a 1,200-character window, which guarantees the target survives tokenizer truncation. This uses a label to construct a training example and never to construct an inference input; Section V-F shows it nevertheless has a cost.

D. Risk Prioritization and Plain-Language Explanation

Each category carries a fixed High/Medium/Low level scored from the weaker party’s perspective (8/22/11) using five criteria: financial exposure, irreversibility, restriction of future freedom, asymmetry, and survival beyond termination. This is category-level prioritization, not clause-level assessment: it cannot distinguish a \$10,000 liability cap from a \$10,000,000 one. A FLAN-T5-small is fine-tuned to map a clause to a plain-English sentence, seeded with 41 author-written category templates.

TABLE I
Presence models, held-out micro-F1

Model	micro-F1	Prec.	Rec.
AND ensemble	0.776	0.772	0.780
TF-IDF baseline	0.775	0.742	0.810
Ensemble AVG	0.718	—	—
Ensemble OR	0.696	—	—
Transformer (max-pool)	0.694	0.562	0.908

IV. Experimental Setup

A. Dataset and Split

All experiments use CUAD v1. The split is drawn at the contract level, 80/20, with a fixed seed: 408 training and 102 test contracts, 20,910 (contract, category) rows at a 32.1% positive rate. The same split is loaded from disk by every model, so “test” denotes the same 102 unseen contracts throughout. Five categories have at most seven positive test contracts.

B. Evaluation Metrics and Uncertainty

Presence is scored by micro-F1 over the 4,182 test decisions. Because the 41 decisions within a contract are not independent, intervals come from a cluster bootstrap that resamples contracts with replacement (10,000 resamples), not individual cells. Span extraction is scored by token-level F1 against the annotated span. Generation is scored by ROUGE-L against two different reference sets, described in Section V-E.

C. Implementation and Reproducibility

Training ran on a single Apple M4 laptop with 16 GB of unified memory using the Metal backend; no CUDA was used at any point. No hyperparameter was tuned. Learning rates, batch sizes and epochs are published defaults; the window geometry follows from a measurement on training data; the decision threshold is the softmax default. This protects the held-out evaluation, but it also means the transformer competes at conventional settings against a baseline with essentially nothing to tune — a limitation revisited in Section V-G. The split, the fitted baseline, the three checkpoints and a standalone evaluation notebook are saved to disk, and a verification script cross-checks all 84 reported figures and 17 derived comparisons against the stored artifacts.

V. Results and Discussion

A. Presence Classification

Table I reports micro-F1 over the 4,182 test decisions. The windowed transformer loses to the lexical baseline by 0.081; the cluster bootstrap places that gap at [+0.063, +0.099], significant in every resample. The AND ensemble finishes nominally first at 0.776, with a margin of +0.0014 and an interval of [−0.0068, +0.0089] — a tie.

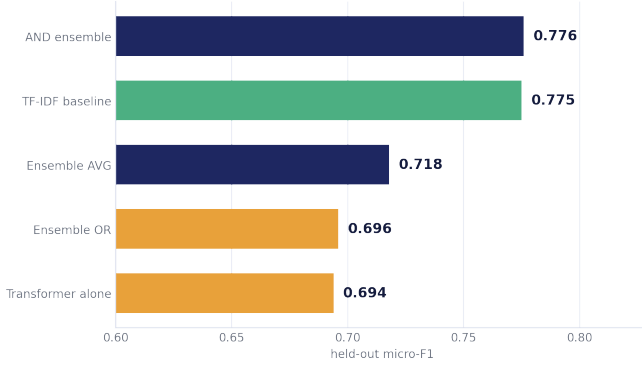


Fig. 1. Presence models on the held-out test set. The lexical baseline and the conjunctive ensemble are statistically indistinguishable; the windowed transformer is not.

TABLE II
Alternative explanations for the transformer’s deficit

Control	Effect on micro-F1	Verdict
All 53,662 windows	$-0.0004 [-0.010, +0.010]$	none
Random seed, 3 runs	SD 0.0085; gap 9.5 SD	none
Aggregation, 6 rules	best $0.750 < 0.775$	none
Threshold, swept	$+0.056$ at $t=0.90$	partial

TABLE III
Uncertainty estimates against the $+0.0014$ ensemble margin

Source	Magnitude	$\times$ margin
Ensemble margin over baseline	0.0014	1.0
Model-seed SD (3 runs, ensemble)	0.0022	1.6
Dataset-split SD (20 splits)	0.0063	4.5
Bootstrap half-width (contracts)	0.0078	5.6

B. Robustness Controls

A bootstrap answers whether a gap could be sampling noise; it cannot answer whether the model was handicapped. Table II reports four controls.

Doubling the training data changed micro-F1 by -0.0004 while raising recall from 0.908 to 0.927 and lowering precision from 0.562 to 0.555: more data made the model fire more readily rather than more accurately, which is what a maximum over approximately 35 windows predicts. Refitting the baseline on 20 independent splits gives a standard deviation of 0.0063, the largest of the three uncertainties in Table III and the one most often omitted, since reseeding holds the test set fixed and cannot observe it.

An ablation over aggregation rules on identical window scores (Table IV) shows that the top-3 mean outperforms the maximum at the default threshold, while the maximum is best once each rule receives its own threshold. No rule at any threshold reaches the baseline, which closes the last available explanation. The threshold control is the one that qualifies the finding: the unchosen 0.5 boundary costs the transformer 0.056, so part of its deficit is an operating-point artefact. We do not adopt the swept threshold, since

TABLE IV
Aggregation ablation on identical window scores

Aggregation	F1@0.5	Best F1	at t
Max (used here)	0.6943	0.7504	0.90
Top-3 mean	0.7144	0.7210	0.40
Top-5 mean	0.6661	0.7192	0.30
Mean over all windows	0.1664	0.7114	0.10
Noisy-OR	0.5897	0.7022	0.95
Second-highest window	0.6832	0.7016	0.30
TF-IDF baseline	0.7747	—	—

TABLE V
Presence performance by risk band (threshold 0.5)

Band	Model	Prec.	Rec.	Missed
High (176 pos.)	Transformer	0.404	0.841	28
	TF-IDF	0.540	0.648	62
	AND ensemble	0.585	0.625	66
	OR ensemble	0.391	0.864	24
Medium (498 pos.)	AND ensemble	0.647	0.673	163
	OR ensemble	0.423	0.922	39
Low (674 pos.)	AND ensemble	0.924	0.901	67
	OR ensemble	0.803	0.969	21

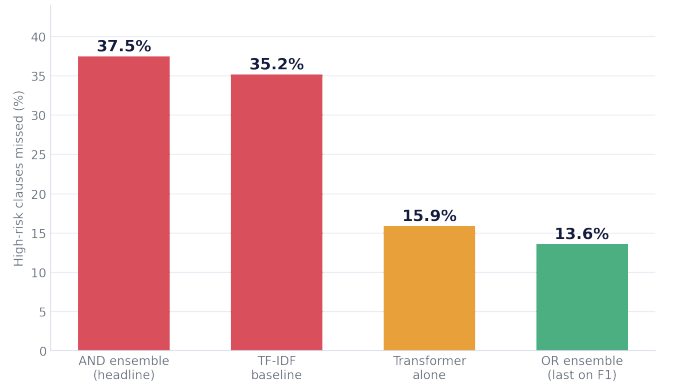


Fig. 2. Percentage of high-risk clauses missed by configuration. The configuration ranked first on aggregate micro-F1 is the worst of the four for the clauses carrying the highest risk.

selecting it on test data would invalidate the held-out evaluation; the appropriate remedy is to carve a validation split from the 408 training contracts, select the threshold there, and fit a calibration map on the same split.

C. Risk-Weighted and Rare-Category Analysis

Micro-F1 weights all 41 categories equally; the system’s purpose does not. Table V scores each model within each risk band.

The configuration ranked first on aggregate micro-F1 misses 37.5% of high-risk clauses; the configuration ranked last misses 13.6%. Two thirds of the aggregate’s mass is low-risk categories such as dates, party names and governing law, so the metric is dominated by the clauses that matter least. On the ten rarest categories the conjunctive

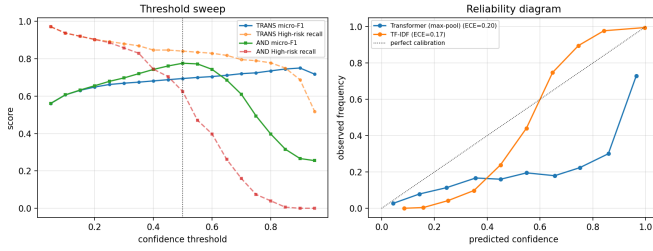


Fig. 3. Threshold sweep (left) and reliability diagram (right). The default 0.5 boundary is a poor operating point for the max-pooled transformer, and the score it thresholds is not calibrated.

TABLE VI
ROUGE-L under two reference sets, same 150 clauses

System	References	ROUGE-L
Fine-tuned FLAN-T5	41 templates	0.775
Fine-tuned FLAN-T5	clause-specific	0.116
Zero-shot FLAN-T5	clause-specific	0.299
Template alone, no model	clause-specific	0.115

rule yields no F1 gain over the transformer alone (0.343 in both cases), purchasing 32.5 points of precision with 25.7 points of recall and reducing detections from 35 of 70 to 17 of 70. We report this as a diagnosis rather than a retuning: selecting the OR rule after observing these test-set results would be test-set selection.

D. Calibration

The max-pooled score has an Expected Calibration Error of 0.201 and a Brier score of 0.194, and is overconfident throughout: predictions above 0.9 correspond to the clause being present 72.8% of the time, and predictions in $[0.5, 0.6]$ to 19.5%. A maximum over dozens of windows is not a probability-preserving operation, and nothing in training asked it to be. Any interface presenting this score as a confidence is therefore misleading its user.

E. Span Extraction and Plain-Language Generation

Given an answer-bearing window, token-F1 is 0.763 with a contract-clustered 95% interval of $[0.740, 0.782]$, and 82.7% of predictions overlap the gold span at $F1 \geq 0.5$. The clustered interval is roughly twice the width obtained by resampling clauses independently, which is the practical cost of ignoring within-contract correlation.

CUAD contains no plain-English rewrites, so the 41 generation targets were author-written and every test clause is scored against one of only 41 unique strings. To separate template reproduction from summarization we scored the same 150 held-out clauses against a second, clause-specific reference set drafted independently of the model (Table VI).

The fine-tuned model emits one of the 41 templates verbatim on 100% of inputs, the correct one 70% of the time, and a constant lookup table matches its score. The task became 41-way classification and the metric did not

TABLE VII
End-to-end attrition over 1,348 gold clauses

Stage	Surviving	Share
Gold clauses	1,348	100.0%
Presence detected	1,052	78.0%
Span overlap ≥ 0.5	293	21.7%
Risk label correct	293	21.7%

register the change. A template-emitting model cannot hallucinate in the usual sense; it fails instead by producing fluent, confident text about a different provision, which is undetectable from the output alone.

F. End-to-End Evaluation

Running all stages in series over the 1,348 gold clauses in the test contracts gives Table VII. A clause survives only if its category is detected, the span extracted from the selected window overlaps the gold clause at token-F1 ≥ 0.5 , and the risk label is correct.

Multiplying the component figures predicts 64%; the pipeline delivers a third of that. Decomposing the loss over the 1,052 detected clauses: the selected window contains the gold text only 68.3% of the time, and even then extraction succeeds 39.7% of the time against 82.7% in isolation. The first cause is a detector reused as a locator, since max-pooling was trained to answer whether a window contains a clause and not which window does. The second is the re-centring used at training time, which is legitimate with respect to leakage but narrows the distribution the model can read; mean token-F1 falls from 0.763 to 0.386. Both are addressable without additional parameters.

G. Limitations

No hyperparameter was tuned, which protects the held-out evaluation but leaves the transformer untuned against a baseline with little to tune. ROUGE-L measures its reference set rather than the task, as Section V-E demonstrates directly, and micro-F1 does not measure user harm, as Section V-C demonstrates. The risk bands are the authors' own construct. All results come from 510 US commercial contracts drawn from SEC filings, in English; no contract outside that corpus has been evaluated. Three seeds is a small sample for a standard deviation, and per-category statistics on the rare tail are not meaningful individually.

VI. Engineering Considerations

Training the three models took approximately 49, 23 and 8 minutes respectively on the target laptop. DeBERTa-v3 produced NaN gradients on the Metal backend and was replaced by DistilBERT; gradient clipping was active on 98.4% of steps, so the nominal learning rate overstates the step size actually taken.

Sparse attention was rejected on cost, and that cost was subsequently measured rather than asserted. Longformer-base at 4,096 tokens requires 14.71 GB for a single training

step at batch size 1 on a 16 GB machine — which fits, contradicting our own earlier claim — but 20.75 GB at batch size 2, where paging makes the step 8.4 times slower for twice the work. At 258 s per step the two-epoch schedule used here would take approximately 136 days against 47 minutes for DistilBERT. On the inference side, scanning one median contract for all 41 categories takes 10.6 s with DistilBERT against 454 s with Longformer.

The deployed prototype runs all inference locally on CPU: a median-length contract takes 27.1 s and the worst case 32.4 s. A 30-window cap bounds latency but reads only the first 45,500 characters, and 38.4% of CUAD contracts exceed that, so clauses beyond the cap cannot be detected — a limitation invisible in every offline metric reported above.

VII. Conclusion and Future Work

On a strict held-out protocol, a full-document lexical baseline beats a windowed transformer at clause presence classification, and the result survives controls for training budget, random seed, aggregation rule and decision threshold. More consequentially, the assembled pipeline retains 21.7% of gold clauses where its component scores predict 64%, and the configuration that maximizes aggregate micro-F1 is the worst available choice for the clauses carrying the highest risk. We suggest that pipeline-level survival and a purpose-weighted metric belong alongside component scores in any system of this kind, and that a margin should be compared against seed and split variance, not only against a bootstrap interval, before being reported as a result.

Future work: carving a validation split so that the decision threshold can be selected and the confidence estimate calibrated without touching test data; a purpose-weighted selection protocol; abstention for the span model through unanswerable training examples; a locator trained for locating rather than a detector reused as one; a clause-diverse generation target set; and evaluation on contracts drawn from outside CUAD.

Acknowledgment

The authors thank Utsha Kumar Roy for requiring a strict held-out evaluation protocol, which shaped how this work reports its results.

References

- [1] D. Hendrycks, C. Burns, A. Chen, and S. Ball, “CUAD: An expert-annotated NLP dataset for legal contract review,” in *Proc. NeurIPS Datasets and Benchmarks*, 2021.
- [2] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, “SQuAD: 100,000+ questions for machine comprehension of text,” in *Proc. EMNLP*, 2016, pp. 2383–2392.
- [3] I. Chalkidis et al., “LEGAL-BERT: The muppets straight out of law school,” in *Findings of EMNLP*, 2020, pp. 2898–2904.
- [4] I. Chalkidis et al., “LexGLUE: A benchmark dataset for legal language understanding in English,” in *Proc. ACL*, 2022.
- [5] I. Beltagy, M. E. Peters, and A. Cohan, “Longformer: The long-document transformer,” *arXiv:2004.05150*, 2020.
- [6] O. Maron and T. Lozano-Pérez, “A framework for multiple-instance learning,” in *Proc. NIPS*, 1998, pp. 570–576.

- [7] C. Raffel et al., “Exploring the limits of transfer learning with a unified text-to-text transformer,” *JMLR*, vol. 21, no. 140, pp. 1–67, 2020.
- [8] H. W. Chung et al., “Scaling instruction-finetuned language models,” *JMLR*, vol. 25, no. 70, pp. 1–53, 2024.
- [9] L. Manor and J. J. Li, “Plain English summarization of contracts,” in *Proc. Natural Legal Language Processing Workshop*, 2019, pp. 1–11.
- [10] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in *Proc. ACL*, 2020, pp. 1906–1919.
- [11] W. Kryściński, B. McCann, C. Xiong, and R. Socher, “Evaluating the factual consistency of abstractive text summarization,” in *Proc. EMNLP*, 2020, pp. 9332–9346.
- [12] C.-Y. Lin, “ROUGE: A package for automatic evaluation of summaries,” in *Text Summarization Branches Out*, 2004, pp. 74–81.
- [13] T. Zhang, V. Kishore, F. Wu, K. Q. Weinberger, and Y. Artzi, “BERTScore: Evaluating text generation with BERT,” in *Proc. ICLR*, 2020.
- [14] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proc. ICML*, 2017, pp. 1321–1330.
- [15] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT,” in *Proc. EMC² Workshop, NeurIPS*, 2019.